

Real-Time FACE DETECTION And RECOGNITION For Security Purposes

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This article demonstrates real-time training, detection and recognition of a human face with OpenCV using Eigenface algorithm.

There are various biometric security methodologies including iris detection, voice, gesture and face recognition, and others. Among these, face recognition plays a vital role and is one of the emerging technologies for security applications. To begin with, we need to understand the logic of training, detection and recognition of human faces.

Face detection

Fortunately, a human face has some easily recognisable features that cameras can lock on to—pair of eyes, nose and mouth. By being able to detect a face in a scene, the camera can concentrate its autofocus on the user's face to ensure it is the primary subject in focus within the image.

Once a face has been detected, the system tracks that face as best it can. If the user turns away, detection lock is lost. But thanks to some camera manufacturers who have enhanced the detection capability of their devices, a lock can be maintained even when the subject has turned to the extent that only one remains visible.

If several faces are detected, not all of them can be in focus. Some face detection modes can lock onto several faces, in which case the primary subject of focus will usually be the closest or the most prominent face. This can be overridden by selecting a face manually.

Face recognition

Face recognition is an enhanced form of face detection. Instead of simply letting the camera recognise any face in the crowd, the camera can be config-



Fig. 1: Face recognition using OpenCV with welcome GUI

ured to recognise a particular person's face and use that as the focal point. You can take photos and save these in the memory. This could be a useful tool when searching for images of a particular person who was recognised by the camera at the time the picture/video was taken.

Face recognition can be implemented using many algorithms like Eigenface, Fisherface, local binary patterns histogram (LBPH) and so on. Eigenface was the first successful technique used for face recognition. It uses principal component analysis (PCA) to project an image to a low-dimension feature space.

Fisherface method is enhancement of Eigenface that uses Fisher's linear discriminant analysis (FLDA).

Eigenface algorithm

In Eigenface algorithm, Eigenface denotes a set of Eigenvectors. These are used in computer vision for human face recognition.

A set of Eigenfaces can be generated by performing a mathematical process of PCA, where it identifies variations in face images in an entire image space as a single point in $n \times n$ -dimensional image space. These

vectors are called Eigenvectors. Since we have not used any high-resolution digital camera to record images in this demonstration, space occupied by every single face is small.

Software

A system loaded with Ubuntu 18.04, or Raspberry Pi installed with NOOBS operating system of any version of Raspbian, or Debian and an external USB camera (optional) can be used. (Testing was done using a laptop with inbuilt camera and Ubuntu 18.04 at EFY Lab.)

To install OpenCV, type the following commands in the terminal:

```
$sudo apt-get update
$sudo apt-get upgrade
$sudo apt-get install python-opencv
$sudo apt-get install libopencv-dev
```

These will install all libraries required for running OpenCV and Python. To check whether OpenCV has been installed, start Python interpreter from the terminal by typing the following:

```
$python
>>> import cv2
>>>
```

If there is no error message, it means OpenCV has been installed successfully.